QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 05

Regional, Site & Urban Climate

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	5.0 - Site Climate (13 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map the 13-slide deck by site-climate logic: regional vs site/microclimate, site-data workflow, primary/secondary/tertiary/local factors, local modifiers, deviation factors and urban climate. Temperature inversion is marked Partial because the deck only lists it as a deviation factor and does not explain the inversion mechanism.

RESULT	Lecture 05 is one of the most exam-stable Section B decks. Site / regional / microclimate factors appear throughout the supplied papers, and the deck also directly supports the recurring site-design process and urban-climate deviation questions. Only temperature inversion and global-scale topography require substantial support from elsewhere.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

The lecture defines regional, site and microclimate, explains why macroclimatic data alone are insufficient, and gives a practical site-climate assessment sequence. It then organizes climate-forming factors into primary, secondary, tertiary and local/minor levels, explains specific modifiers such as soil, topography, windbreakers, ocean currents, smog/fog and sand storms, lists deviation variables, and closes with human causes and measured effects of urban climate. This structure maps exceptionally well to the historical question bank.

13	23	18	1	4	288
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
2-4	Regional vs site/microclimate + factor hierarchy	2016-2024 repeated site-factor family	Highest-value definition/factor block.
3	Site-climate data / large-site design workflow	2020, 2022, 2023	Direct repeated design-process block.
5-9	Primary-secondary-tertiary-local modifiers	2016, 2017, 2018, 2020, 2021, 2024	Core long-answer + short-note material.
10	Deviation factors incl. temperature inversion	2018/2021 inversion questions partial	Useful checklist; inversion mechanism not developed.
11-12	Urban climate causes + effects	2021, 2022, 2023, 2024	Strong recent four-paper family.

Priority note If time is short, study Slides 2-4, 7-9 and 11-12. The exam repeatedly asks the same chain: regional data -> local modifiers -> site climate -> design decision.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q4(a)	25	1	Identify local factors responsible for developing a site/microclimate and distinguish it from regional or macroclimate.	Exact	2-9	The definition, regional/site distinction and local-factor hierarchy are directly supplied.
2017	Q3(a)	15	2	Differentiate Site Climate from Regional Climate.	Exact	2	Direct definition/comparison match.
2017	Q4(a)	10	2	What is microclimate? What do you know about microclimate factors?	Exact	2, 4-9	Microclimate is defined and the factor hierarchy/local modifiers are developed.
2017	Q4(b)	15	2	What is site climate? Which factors are responsible for site climate?	Exact	2, 4-9	Direct lecture match.
2017	Q4(c)	10	2	Write short notes on Ocean Current and Natural Wind Breaker.	Exact	7-8	Both modifiers are explicitly explained.
2017	Q7(b)	10	2	What are the tertiary factors constituting regional climate?	Exact	4, 7-9	The tertiary-factor list and examples are directly provided.
2018	Q6(a)	20	4	What is site climate? Elaborate the factors of site climate.	Exact	2, 4-9	Direct definition + factor hierarchy match.
2018	Q6(b)	15	4	What do you understand by temperature inversion? Discuss with example.	Partial	10	Temperature inversion is only named as a deviation factor; definition, mechanism and examples are absent.
2020	Q3(a)	10	5	Describe the fundamental process of designing a large site using site-climate knowledge.	Exact	3	The slide gives the practical sequence: start with regional data, assess deviations, use local observation/expert advice and identify dominant local factors.
2020	Q3(b)	20	5	Explain five factors that contribute to formation of site climate.	Exact	4-9	The deck supplies more than five factors with mechanisms.
2020	Q6(b)	15	5	Identify local factors responsible for developing a site/microclimate.	Exact	4-9	Direct local-modifier match.
2021	Q1(a)	10	6	Assess the influence of topography from the perspective of global climate variation.	Partial	7	The slide explains local topographic effects, not the full global-scale topography mechanism.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q3(a)	10	6	Distinguish between urban climate and macroclimate.	Direct	2, 11-12	Macro/site framework plus urban causes/effects allows a clear comparison; the student must synthesize it.
2021	Q3(b)	10	6	Explain temperature inversion from the perspective of global warming.	Partial	10	Only the term is listed; the inversion mechanism and global-warming discussion are not taught.
2021	Q4(b)	10	6	Explain site climate and factors responsible for it.	Exact	2, 4-9	Direct match.
2022	Q4(a)	10	8	Distinguish between site climate and regional climate.	Exact	2	Direct match.
2022	Q4(b)	10	8	Describe the fundamental process of designing a site using site-climate knowledge.	Exact	3	Direct site-data/design-process match.
2022	Q4(c)	15	8	Analyse the factors responsible for deviation of urban climate and explain their effects.	Exact	11-12	Human interventions and three quantified/qualitative urban effects are directly given.
2023	Q3(b)	10	10	Describe the fundamental process of designing a residential building/site in KUET using site-climate knowledge.	Exact	2-3	Use regional data as baseline, assess local deviations, identify dominant factors and select the suitable habitation area.
2023	Q3(c)	15	10	Analyse factors responsible for deviation of urban climate and their effects.	Exact	11-12	Direct repeated urban-climate match.
2023	Q4(a)	5	10	Write down topographical factors responsible for global climate variation.	Partial	7	Topography is discussed as a local/tertiary modifier; full global topographic processes require Lecture 02/reference material.
2024	Q3(a)	10	13	What is site climate, and what factors cause deviations in local climate?	Exact	2, 4-10	Direct match to site-climate definition, modifiers and deviation-factor checklist.
2024	Q3(b)	8	13	Urban climate depends on human intervention - explain.	Exact	11-12	Changed surfaces, buildings, anthropogenic heat and pollution are directly listed with effects.

Coverage key	Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

This lecture rewards structure. Most questions can be answered by moving from scale to mechanism: regional/macro climate -> local modifiers -> site/microclimate -> urban/human deviation. The factor hierarchy is repeated enough that it should be memorized as a table, not as prose.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Site / microclimate factors + local deviation	2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024	2, 4-10	Appears across all supplied papers in direct or close form.
S	Site vs regional / macro climate	2016, 2017, 2021, 2022	2	Stable definition/comparison family.
S	Urban climate: human intervention + deviation/effects	2021, 2022, 2023, 2024	11-12	Four recent consecutive papers.
A	Fundamental site-design process	2020, 2022, 2023	3	Repeated 10-mark workflow question.
A	Local modifiers / tertiary factors / ocean current / windbreaker	2017, 2020 (+ broader repeats)	4-9	Easy marks through organized factor table.
B	Temperature inversion	2018, 2021	10 only	Question repeats, but this deck is only a partial source.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“site climate / microclimate / regional climate”	2	Macro data are a guide; site climate is the actual project-area climate in horizontal extent + height.
“design a large site / KUET site”	3	Start regional data -> assess deviations -> consult/observe -> identify local factors -> choose habitation zone.
“primary / secondary / tertiary / local factors”	4-9	Primary: global/temperature-humidity-airflow; Secondary: location-diurnal-altitude; Tertiary: soil-topography-windbreakers-water-cities-desert; Local: ocean current-sky-smog/fog/storms.
“urban climate / human intervention”	11-12	Changed surfaces + buildings + energy seepage + pollution -> hotter, drier, lower wind, turbulence.
“temperature inversion”	10 + other source	Only the term is listed here; do not fabricate its mechanism from this slide.
“ocean current / natural windbreaker”	7-8	Windbreaker modifies flow; ocean current transports heat and changes coastal temperature/humidity/rainfall.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2	Regional / macro vs micro / site climate	2016 Q4(a); 2017 Q3(a), Q4(a,b); 2022 Q4(a); 2024 Q3(a)	Memorize definition + 4-point comparison.
3	Site climate data and design process	2020 Q3(a); 2022 Q4(b); 2023 Q3(b)	Turn the bullets into a numbered workflow.
4-6	Factor hierarchy + primary/secondary factors	2016-2024 site-factor family	Memorize categories before details.
7-9	Tertiary/local modifiers: soil, topography, windbreakers, ocean current, smog/fog/sand	2017 Q4(c), Q7(b); 2020 Q3(b), Q6(b)	Prepare as factor -> climatic effect table.
10	Deviation factors	2018/2021 inversion partial; 2024 local deviation	Use as checklist; inversion needs another source.
11-12	Urban climate causes and effects	2021 Q3(a); 2022 Q4(c); 2023 Q3(c); 2024 Q3(b)	High priority recent block; learn the reported 8°C, 5-10% RH and less-than-half wind facts as lecture values.
13	Closing	None	Skip.

6. Coverage Gaps & Diagnostic Exclusions

The deck is unusually complete for site/urban climate, but two exam families extend beyond it: temperature inversion and global-scale topographic climate variation. Those should not be answered from Slide 10 or Slide 7 alone.

Gap	Affected questions	Action
Temperature inversion mechanism/example is absent	2018 Q6(b); 2021 Q3(b)	Use the reference/class note; Slide 10 only proves the topic belongs to deviation factors.
Global-scale topography is not fully developed	2021 Q1(a); 2023 Q4(a)	Use Lecture 02 topography/global-wind material plus reference explanations.
Some direct site factors such as vegetation are not separately expanded	Broad site-climate answers	Use the factor hierarchy as primary source; add vegetation only from another supplied source if needed.
Urban-climate values are lecture-specific examples	Urban deviation questions	Present them as course-slide values rather than universal constants.

Mapping discipline The report preserves the lecture's hierarchy and wording even where categories overlap. It also keeps temperature inversion as Partial because the source deck merely lists it; the mechanism is not silently supplied from general knowledge.

7. Recommended Study Plan

This deck should be studied as one logic chain. First learn the scale distinction, then the factor hierarchy, then the design process, and finally urban deviation. That order mirrors the repeated exam pattern.

Step	Study block	Slides	Practice immediately
1	Site vs regional / microclimate definition	2	2016 Q4(a), 2017 Q3(a), 2022 Q4(a).
2	Factor hierarchy + modifiers	4-9	2017 Q4(b,c), Q7(b), 2020 Q3(b).
3	Site-design process	3	2020 Q3(a), 2022 Q4(b), 2023 Q3(b).
4	Urban climate	11-12	2022 Q4(c), 2023 Q3(c), 2024 Q3(b).
5	Gap check: inversion / global topography	7, 10	Open the supporting lecture/reference before attempting those questions.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Regional vs site comparison	Use scale, data source, local modifiers, design usefulness.
A	Factor hierarchy table	Primary / Secondary / Tertiary / Local-minor with examples.
A	Site-design flow	Regional data -> deviation -> local factors -> suitable zone -> design response.
A	Urban-climate cause/effect table	Cause in left column, temperature/RH/wind/pollution effect in right.

Bottom line	Lecture 05 is a must-prepare Section B lecture. It directly covers the most persistent site-climate family and the recent urban-climate trend; only inversion and global topography need meaningful supplementation.